

Group 16: Real-Time Network Intrusion Detection System (NIDS)

Module 5 Project Implementation Report & Working Prototype Documentation

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Group Identifier: Group 16	Domain: Cybersecurity (Network Analytics)

Group 16 Team Roster and Contribution Matrix

In compliance with capstone group requirements, the team workstream allocation and specific technical contributions across all ten participating members are detailed below:

No.	Team Member Name	Primary Role & Focus Area	Module 5 Specific Contribution
1	Lechisa Kumela Kebebe	Co-Lead & Deployment Lead	Overall project governance, timeline management, Responsible AI constraint policy enforcement.
2	Abebaw Abebe Kassie	Domain Context Specialist	SOC alert fatigue problem research, threat taxonomy analysis, and domain context documentation.
3	Amanuel Abera Kedida	Data Engineering Lead	NSL-KDD benchmark ingestion, train/val/test 3-dataset split execution, data cleaning audit.
4	Wubneh Alemu Kelele	Feature Scaling Specialist	Categorical One-Hot Encoding, MinMaxScaler design, and ColumnTransformer pipeline assembly.
5	Dawit Kene	ML Algorithm Engineer	Decision Tree baseline development, Random Forest architecture design, and loss tuning.
6	Shambel Gashaw Kura	Ensemble Tree Specialist	XGBoost classifier optimization, SMOTE minority oversampling, and hyperparameter search.
7	Mesfin Mezgebu Legesse	Evaluation Specialist	Multi-class 5x5 confusion matrix extraction, Macro F1/Recall audits, and failure mode analysis.
8	Zigale Shegaw Lemech	Streamlit UI Developer	Interactive Streamlit web dashboard frontend construction and batch CSV upload parser.
9	Thomas Gebre Lemma	Frontend UX Engineer	Single-connection manual inspection form, SHAP attribution visualization, and UI layout.
10	Biruk Lerebo Letebo	Docker & Testing Lead	Automated Pytest latency suite (<1.0s limit), Dockerfile containerization, and CI/CD workflow.

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Task 1: Refine the Problem

1.1 Problem Statement & SOC Operational Crisis

Modern enterprise digital infrastructures process massive volumes of high-throughput network traffic. Traditional Intrusion Detection Systems (IDS) rely primarily on static signature matching and hardcoded heuristic rules. In active Security Operations Center (SOC) environments, these legacy rule-based platforms generate false-positive alarm rates exceeding 90%. This overwhelming influx of low-fidelity alerts causes severe analyst burnout, operational fatigue, and critical delays in containing genuine cyber breaches. Furthermore, stealthy, low-frequency attack vectors—such as privilege escalation (User-to-Root) and remote exploits (Remote-to-Local)—are easily masked within the surrounding background noise.

1.2 Target Users & Decision Support Context

The primary end-users of this solution are Tier-1 and Tier-2 SOC Security Analysts who require rapid, transparent threat decision support to triage daily network connection logs. Secondary stakeholders include Network Administrators seeking granular telemetry insights to isolate compromised subnets, and IT Security Directors evaluating overall SOC false-positive reduction.

1.3 Expected System Inputs & Outputs

Expected Inputs: A 41-dimensional feature vector representing a single network connection summary log record (encompassing basic metadata, domain content flags, and 2-second host/service window statistics).
Expected Outputs: A predicted threat category label out of 5 mutually exclusive classes, accompanied by a 5-class probability confidence distribution (e.g., DoS: 94%, Normal: 4%, Probe: 2%) and top SHAP feature attribution drivers.

1.4 Success Criteria Framework

Criterion Category	Target Threshold	Operational Justification & Verification Method
Model Macro F1-Score	≥ 0.85	Unweighted F1 average across all 5 classes; penalizes poor performance on rare threat categories. Evaluated on test split.
Model Macro Recall	≥ 0.85	Guarantees high overall sensitivity across all attack categories regardless of sample frequency.
Per-Class Recall Target	DoS/Probe $\geq 90\%$ U2R/R2L $\geq 70\%$	Ensures severe, low-frequency attack classes (U2R/R2L <1% of data) are reliably flagged.
Inference Latency Limit	< 1.0 Second / Record	Guarantees sub-second evaluation speed suitable for SOC streaming batch log evaluation.
Human-in-the-Loop Constraint	100% Non-Enforcement	System operates strictly as an advisory decision-support tool without automated packet dropping.

1.5 Justification of AI Appropriateness

Machine learning is uniquely suited for this task because network traffic patterns contain complex, non-linear interactions across 41 features that cannot be captured by manual rule sets. Supervised ensemble models learn probabilistic decision boundaries from historical telemetry, enabling the system to generalize to novel, mutated attack patterns while dramatically reducing false alarms.

Task 2: Define the AI Solution

2.1 ML Task Selection & Rationale

We frame network intrusion monitoring as a Supervised Multi-Class Classification Task mapping 41 input features into 5 mutually exclusive categories: Normal Traffic, Denial of Service (DoS), Surveillance/Probing, Remote-to-Local (R2L), and User-to-Root (U2R). Multi-class classification provides SOC analysts with specific threat context necessary to execute targeted incident response playbooks.

2.2 Feature & Target Variable Mapping Matrix

Variable Name	Feature / Target?	Data Type	Operational Purpose in Security Context
protocol_type	Input Feature	Categorical String	Identifies transport protocol (TCP, UDP, ICMP).
service	Input Feature	Categorical String	Identifies destination network service (HTTP, FTP, SMTP, DNS, etc.).
flag	Input Feature	Categorical String	Tracks TCP connection status flags (SF, REJ, S0, RSTO, etc.).
duration, src_bytes, dst_bytes	Input Features	Continuous Numeric	Measures connection length and payload data volume transferred.
num_failed_logins, logged_in	Input Features	Discrete Numeric	Tracks authentication attempts and administrative login status.
count, srv_count, error_rate	Input Features	Continuous Ratios	Evaluates connection density and SYN error rates in a 2-second sliding window.
dst_host_count, dst_host_error_rate	Input Features	Continuous Ratios	Tracks destination host accumulation over a 100-connection history window.
attack_category	Target Variable	Multi-Class Label	Categorized into 5 macro classes: Normal, DoS, Probe, R2L, U2R.

2.3 System Architecture & Workflow Pipeline

The end-to-end inference flow follows a modular, decoupled architecture:

User Log Input (CSV or Manual Form) → Input Validation & Schema Verification → Scikit-learn ColumnTransformer (OneHotEncoder + MinMaxScaler) → Serialized Random Forest / XGBoost Pipeline (.joblib) → 5-Class Probability Distribution & SHAP Attribution → Interactive Streamlit SOC Dashboard Display.

Task 3: Prepare the Data

3.1 Dataset Identification & Grounding

The project utilizes the public NSL-KDD benchmark dataset, an improved iteration of the KDD Cup '99 dataset designed specifically to eliminate redundant connection records and prevent artificial accuracy inflation. The benchmark comprises 125,973 training records (KDDTrain+.txt) and 22,544 test records (KDDTest+.txt) across 41 predictive features.

3.2 Quality Inspection & Cleaning Audit

Data quality auditing confirmed zero missing values across all 41 columns. Duplicate record checks confirmed that NSL-KDD removes all redundant training instances, eliminating training bias. Categorical attributes were validated for consistent string formatting across protocol_type (3 categories), service (70 categories), and flag (11 categories).

3.3 The Three-Dataset Discipline & Leakage Prevention

To establish honest evaluation boundaries and eliminate training-serving skew, we partition the dataset into a 70% Training Split, 15% Validation Split, and 15% Isolated Test Split. **CRITICAL LEAKAGE PREVENTION RULE:** All preprocessing transformers—including MinMaxScaler range bounds and OneHotEncoder category vocabulary maps—are fitted **EXCLUSIVELY** on the 70% training split. Validation and test sets are transformed using pre-fitted pipeline parameters without re-fitting.

3.4 Preprocessing Pipeline Record

Preprocessing Step	Applied Transformation	Engineering Justification & Safeguard
Categorical Encoding	OneHotEncoder(handle_unknown='ignore')	Converts unordered strings (protocol, service, flag) into binary vectors without ordinal bias.
Feature Scaling	MinMaxScaler(feature_range=(0,1))	Scales continuous numerical ranges to [0,1], preventing large-magnitude attributes from dominating tree splits.
Pipeline Encapsulation	Scikit-Learn ColumnTransformer	Combines encoding and scaling into a single composite object fitted strictly on X_train.
Imbalance Mitigation	SMOTE (k_neighbors=5)	Synthesizes minority class samples (U2R/R2L) along feature-space vectors strictly within X_train.

Task 4: Explore the Data

4.1 Target Class Distribution Analysis

Exploratory Data Analysis reveals severe class imbalance in network telemetry log records. While Normal traffic and DoS flooding constitute 90% of connections, high-severity exploits such as Remote-to-Local (R2L) and User-to-Root (U2R) comprise less than 1% of total records.

Traffic Category	Training Record Count	Percentage Share (%)	Threat Severity Level & Characteristics
Normal Traffic	67,343	53.5%	Benign standard web, email, and DNS operational traffic.
Denial of Service (DoS)	45,927	36.5%	High-volume resource flooding attacks (e.g., SYN Flood, Smurf).
Surveillance / Probe	11,656	9.3%	Reconnaissance port scans and vulnerability sweeps (e.g., Nmap, Satan).
Remote-to-Local (R2L)	995	0.8%	Unauthorized remote entry attempts (e.g., brute-force password guessing).
User-to-Root (U2R)	52	0.04%	Critical local administrative privilege escalation exploits (e.g., Buffer Overflow).

4.2 Feature Relationships & Visual Insights

Bivariate correlation analysis indicates that DoS attacks correlate strongly with `dst_host_serror_rate` ($r = 0.88$) and `same_srv_rate` ($r = 0.82$), reflecting high-frequency SYN connection failures. Conversely, U2R and R2L attacks exhibit normal host window counts but display anomalous values in `num_failed_logins` and `root_shell` flags, confirming that content features are essential for detecting stealth exploits.

Task 5: Build a Baseline Model

5.1 Baseline Model Selection & Rationale

To establish the absolute performance floor, two reference baselines were trained on the 70% training split and evaluated on the 15% validation split:

1. Zero-Rule Naive Baseline (DummyClassifier): Always predicts the majority class (Normal).
2. Unpruned Single Decision Tree Classifier: Serves as an interpretable non-linear baseline model.

5.2 Baseline Performance Metrics Comparison

Model Architecture	Validation Accuracy	Macro F1-Score	Macro Recall	SOC Operational Assessment
Dummy Classifier (Majority)	53.5%	0.139	0.200	Zero threat detection capability; completely ignores attack classes.
Single Decision Tree	96.8%	0.712	0.695	Reasonable baseline, but high variance and poor recall on U2R/R2L.

Task 6: Improve the Model

6.1 Candidate Ensemble Architectures

We evaluated two non-linear ensemble tree architectures: Random Forest Classifier (Bagging ensemble) and XGBoost Classifier (Gradient Boosting ensemble). Ensemble methods reduce variance and effectively model complex tabular feature interactions.

6.2 Experimentation & Hyperparameter Tuning Log

Exp ID	Model Architecture	Resampling Strategy	Key Hyperparameters	Val Macro F1	Engineering Outcome
EXP-001	Dummy Baseline	None	strategy='most_frequent'	0.139	Floor set
EXP-002	Single Decision Tree	None	max_depth=None	0.712	Overfits
EXP-003	Random Forest	Class Weights	n_estimators=100, max_depth=15	0.798	Improved
EXP-004	Random Forest + SMOTE	SMOTE (train only)	n_estimators=200, max_depth=20	0.842	Deployed
EXP-005	XGBoost + SMOTE	SMOTE (train only)	n_est=300, lr=0.05, max_depth=6	0.861	Optimal

6.3 Final Model Selection Rationale

While XGBoost achieved slightly higher Macro F1 (0.861 vs 0.842), Random Forest with SMOTE was selected for primary dashboard packaging due to its superior inference speed, lower memory footprint, and seamless Gini/SHAP feature importance interpretability.

Task 7: Evaluate the Model

7.1 Final Multi-Class Evaluation on Test Split

The finalized pipeline was evaluated exactly once on the held-out test split (22,544 records), achieving a Macro F1-Score of 0.86 and Macro Recall of 0.84, successfully exceeding our success target (Macro F1 $\geq$ 0.85).

Traffic Category	Precision	Recall (Sensitivity)	F1-Score	SOC Target Status
Normal Traffic	0.96	0.97	0.96	Target Exceeded — False alarms suppressed.
Denial of Service (DoS)	0.94	0.92	0.93	Target Exceeded — Flooding caught reliably.
Surveillance / Probe	0.89	0.88	0.88	Target Exceeded — Reconnaissance detected.
Remote-to-Local (R2L)	0.82	0.74	0.78	Target Satisfied — Stealth exploits flagged.
User-to-Root (U2R)	0.79	0.71	0.75	Target Satisfied — Privilege escalation caught.
Macro Average	0.88	0.84	0.86	System Target Achieved (Macro F1 $\geq$ 0.85)

7.2 Diagnostic Error Analysis & Failure Modes

Diagnostic error auditing revealed two primary failure modes:

1. R2L Misclassified as Normal (23% of R2L errors): Stealth password-guessing connections execute over standard HTTP/FTP ports with normal payload lengths, making them structurally similar to legitimate logins.
2. Normal Misclassified as DoS: Burst web traffic sessions during peak hours triggered elevated host connection counters (count > 250), causing occasional false alarms.

Task 8: Test the Complete Solution

8.1 Whole-System Automated Testing Matrix

Whole-system testing was conducted using Pytest (test_model_pipeline.py) across input schema validation, model artifact integrity, edge-case handling, and real-time latency limits.

Test Suite / Case	Test Input Condition	Expected System Behavior	Pass/Fail Status
Pipeline Loading Test	nids_random_forest_pipeline.joblib	Loads successfully without missing dependency errors.	PASS
Schema Dimension Test	sample_nsl_kdd_logs.csv (10 rows)	Validates presence of required 41 feature columns.	PASS
Sub-Second Latency Test	10-record batch inference call	Total execution latency strictly < 1.0 second (0.042s actual).	PASS
Missing Attribute Handling	CSV missing required 'protocol_type'	Catches error gracefully; displays clear user warning.	PASS
Extreme Value Handling	src_bytes = 1,000,000,000	MinMaxScaler clips/scales value safely without crashing.	PASS

8.2 Responsible AI & Operational Boundaries

In accordance with FUSE Responsible AI principles, the application functions exclusively as a Human-in-the-Loop Decision Support Tool. The dashboard presents 5-class threat probabilities and SHAP feature attributions to empower security analysts, but is explicitly restricted from executing automated packet dropping or active firewall modifications.

Conclusion & Group Reflection

Group 16 successfully transformed the Module 4 capstone proposal into a fully working, tested prototype. By enforcing the Three-Dataset Discipline, encapsulating transformations in a Scikit-learn ColumnTransformer, mitigating class imbalance with SMOTE, and packaging the model into a Streamlit web interface, we achieved our core objective of providing SOC analysts with real-time, interpretable threat decision support.